

ELANTHIRAYAN MADHAVAN

Senior AI Engineer

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PROFESSIONAL SUMMARY

Senior AI Engineer with 11+ years of experience translating business opportunities into practical, scalable technology solutions. Strong at selecting the right architecture for each problem, Balancing business value, technical feasibility, and long-term maintainability rather than chasing technology trends. Proven ability to design and deliver production-ready enterprise AI systems and cloud-native applications that solve real business problems. Active open-source contributor and creator of AI Concepts Playground with strong connection to the broader AI developer community.

PROFESSIONAL EXPERIENCE

Shell India Markets Pvt Ltd | 2018 - Present

L&T Technology Services (2018–2023), Randstad India Pvt Ltd (2023–2025), Shell India Markets Pvt Ltd – Direct Employee (2025–Present)

Innovation Analyst (AI, HPC & Spatial Computing)

Generated an estimated USD 18 million uplift through safety and operational efficiency of oil and gas assets by leveraging the appropriate emerging digital technologies including AI, gen AI, HPC and spatial computing.

Key Highlights

- Delivered enterprise AI solutions for engineering and technical knowledge management, including Retrieval-Augmented Generation (RAG), intelligent document processing, semantic search, and multi-agent workflows that improved access to enterprise knowledge.
- Designed scalable cloud-native architectures and production-ready applications using Python, FastAPI, Microsoft Azure, and Azure OpenAI, enabling secure and maintainable enterprise AI solutions.
- Improved large-scale document processing by leveraging High-Performance Computing (HPC) to parallelize extraction, preprocessing, embedding generation, and vector indexing for enterprise-scale workloads.
- Led the development of Digital Twin and Spatial Computing solutions, including OpenUSD adoption strategies, immersive 3D visualization platforms, and Apple Vision Pro applications for engineering and hazard awareness.
- Evaluated emerging AI, XR, and cloud technologies through technical research, proof-of-concepts, and internal publications, helping shape enterprise innovation and future technology roadmaps.
- Worked closely with business stakeholders, global engineering teams, and technology partners to define solution architectures, evaluate technologies, and successfully deliver enterprise digital transformation initiatives.
- Established reusable engineering frameworks and technical best practices that accelerated AI solution development across multiple enterprise initiatives.

ASM Technologies Ltd | 2017 - 2018

Full-Stack Software Engineer

Developed enterprise web applications, backend services, and immersive XR training solutions for automotive and semiconductor industries, integrating real-time visualization with cloud-based business systems.

Key Contributions

- Developed immersive VR training applications using Unity for automotive and semiconductor assembly/disassembly processes, reducing training time by 65% compared to conventional methods.
- Built interactive ADAS simulation environments, enabling engineers to visualize, configure, and evaluate vehicle systems from multiple perspectives, improving technical understanding by 40%.
- Designed and implemented REST APIs connecting XR applications with enterprise backend systems, enabling secure data exchange and real-time synchronization.
- Developed web-based administration portals for content management, user analytics, and training reporting using modern web technologies.
- Collaborated with cross-functional engineering teams to deliver enterprise XR solutions for manufacturing and industrial training.

Full-Stack Software Engineer

Designed and developed full-stack web applications, real-time collaboration platforms, and cloud-hosted enterprise solutions across education and event management domains.

Key Contributions

- Developed full-stack features for Skilladda and Xucore e-learning platforms, contributing across frontend, backend, database, and deployment.
- Designed and developed a real-time multiplayer event management platform using Node.js, WebRTC, and MySQL, enabling live audience participation at NVIDIA Gamer Connect events across Asia with 95% live quiz participation.
- Built hybrid mobile applications for Android and iOS using Apache Cordova, extending platform accessibility across mobile devices.
- Managed cloud deployments using Microsoft Azure App Services and Azure Blob Storage, supporting scalable application hosting and content delivery.
- Collaborated with stakeholders throughout the software development lifecycle, delivering production-ready web applications using modern full-stack development practices. Technologies: Node.js, PHP, JavaScript, MySQL, Apache Cordova, Microsoft Azure

OPEN SOURCE CONTRIBUTIONS

AI Concepts Playground (*Creator & Maintainer*) - Open-source interactive AI learning platform with 50+ hands-on lessons covering LLMs, RAG, MCP, AI Agents, and LLM Architecture. Designed interactive visualizations and browser-based experiments to simplify complex AI concepts for developers. <https://imelanthirayan.github.io/aiplayground/>

MilvusLite Kit - Built a lightweight Python toolkit simplifying vector database integration for Retrieval-Augmented Generation (RAG) applications.

Chat UI - Developed a configurable AI chat platform supporting multiple LLM providers and extensible backend integrations.

FolderWatcher - Event-driven Python framework that monitors filesystem changes and automates document ingestion workflows for AI and RAG pipelines.

FileTracker - Python utility for tracking file changes and triggering automated document processing workflows to support enterprise AI and knowledge management solutions.

EXTERNAL PUBLICATIONS

- Presented a peer-reviewed research paper at IIM Bangalore on Reinforcement Learning, developing a Unity-based 3D simulation environment where a robotic arm learned autonomous object grasping and pick-and-place tasks through reinforcement learning.

EDUCATION

M.Tech in Data Science and Machine Learning | 2022

PES UNIVERSITY

Master of Computer Applications | 2018

ANNAMALAI UNIVERSITY

TECHNICAL SKILLS

- Generative AI & Agentic AI: LLMs, Retrieval-Augmented Generation (RAG), LangGraph, AI Agents, Azure AI Foundry, Azure OpenAI, Knowledge Graphs, Neo4j
- Cloud & DevOps: Microsoft Azure, Azure Storage Accounts, Docker, Azure App Services, GitHub Actions, CI/CD, High-Performance Computing (HPC)
- Programming: Python, C#, JavaScript, SQL
- Frameworks: FastAPI, React, Next.js, Node.js
- Spatial Computing: OpenUSD, Digital Twins, Unity, Apple Vision Pro, XR